



SAJJAN SHOPPING

Product Requirements Document (PRD)

Java Command Line Interface (CLI) Shopping Application

Project Name	Sajjan Shopping
Platform	Java Command Line Interface
Application Type	Console-based Shopping Application
Primary Sections	Men and Women
Users	User 1 and User 2
Core Operations	Place, Modify, View and Cancel Order
Technology	Java, OOP, Scanner, ArrayList
Version	1.0

1. Executive Summary

Sajjan Shopping is a beginner-friendly command line shopping application developed in Java. The system provides a simple shopping experience through a menu-driven console interface. Users can choose a Men or Women section, select products, place an order, modify an order, view existing orders and cancel an active order.

2. Problem Statement

Traditional beginner shopping projects often focus only on displaying products. Sajjan Shopping extends this idea by introducing a complete basic order lifecycle. The application is intended to demonstrate how Java can be used to model users, products and orders while providing useful CRUD-like order operations through a CLI.

3. Product Goals

Goal	Description
Simple shopping	Allow users to browse products by category.
Order management	Support placing, modifying, viewing and cancelling orders.
Multiple users	Provide separate sessions for User 1 and User 2.
Java learning	Demonstrate classes, objects, ArrayList, methods and Scanner.
Easy operation	Use clear numbered menus suitable for a command line environment.

4. Target Users

The primary users are customers represented by User 1 and User 2. The project is also designed for students and beginners learning Java programming, object-oriented programming and menu-driven application design.

5. Functional Requirements

ID	Requirement	Expected Behaviour
FR-01	User Selection	User can select User 1 or User 2.
FR-02	Men Section	Display Men products and prices.
FR-03	Women Section	Display Women products and prices.
FR-04	Place Order	Selected product is added and a unique order ID is generated.
FR-05	Modify Order	User can replace/update the selected item before finalising.
FR-06	View Order	Show order ID, section, items, total and status.
FR-07	Cancel Order	Active order can be cancelled by its owner.
FR-08	Logout	Return the user to the main user-selection screen.
FR-09	Exit	Close the application safely.

6. Product Catalogue

Section	Sample Products	Price Range
Men	Shirt, Jeans, Shoes, Watch, Wallet	■499 – ■1,499
Women	Dress, Top, Saree, Shoes, Handbag	■699 – ■1,999

7. User Flow

1. Launch application
2. Select User 1 or User 2
3. Select Men or Women section
4. Browse products
5. Select a product
6. Place or modify the order
7. View order details and status
8. Cancel the order when required
9. Logout or exit

8. Non-Functional Requirements

Category	Requirement
Usability	Menus and prompts should be simple and clearly numbered.
Reliability	Invalid menu selections should not crash the application.
Performance	Operations should respond immediately for the small in-memory dataset.
Maintainability	Products and orders should be represented using separate Java classes.
Portability	Application should run on systems with a compatible Java JDK.

9. Technical Design

Main Java Components

Class / Component	Responsibility
Product	Stores product ID, product name and price.
Order	Stores order ID, user, section, products and order status.
SajjanShopping	Controls menus, user sessions and shopping operations.
ArrayList<Order>	Stores orders during application execution.
Scanner	Reads user input from the command line.

10. Order Lifecycle

Created → Confirmed → Cancelled

When the user places an order, the system generates a unique order number such as ORD101. The order starts with CONFIRMED status. If the user cancels an active order, its status changes to CANCELLED. The order remains available for viewing during the current application session.

11. Validation and Edge Cases

- Invalid user selection → display an error and return to the menu.
- Invalid product number → display an error and ask again.
- No existing orders → display 'No orders found.'
- Unknown order ID → display 'Order not found.'
- Already cancelled order → prevent repeated cancellation.
- Exit option → terminate the program cleanly.

12. Future Enhancements

- Add quantity support and shopping cart functionality.
- Add customer name, phone number and delivery address.
- Add payment methods such as Cash on Delivery and UPI.
- Add search and product filtering.
- Add discount coupons and GST calculation.
- Store orders permanently using MySQL or another database.
- Convert the CLI application into a Java Swing or JavaFX GUI.
- Add administrator login for product and order management.

13. Acceptance Criteria

Test	Pass Condition
User login	User 1 and User 2 can enter their own shopping sessions.
Category	Men and Women products are displayed correctly.
Place	A valid product creates an order with a unique order ID.
Modify	The selected order item can be changed successfully.
View	Order details and total amount are displayed.
Cancel	A confirmed order changes to CANCELLED.
Invalid input	Invalid choices do not terminate the application unexpectedly.

14. Conclusion

Sajjan Shopping provides a clear and practical demonstration of a Java-based shopping system. Its menu-driven design makes the application easy to operate, while its Product and Order classes demonstrate core object-oriented programming concepts. The project can later be extended into a database-backed or graphical e-commerce application.

SAJJAN SHOPPING — Shop Smart, Live Better!